

OTTUMWA, IA, USA

WMO: 725465

Lat: 41.108N

Lon: 92.447W

Elev: 257

StdP: 98.28

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.2	(c) -17.4	(d) -24.4	(e) 0.4	(f) -19.6	(g) -22.1	(h) 0.5	(i) -16.9	(j) 14.0	(k) -6.4	(l) 12.6	(m) -7.4	(n) 4.5	(o) 300	(p) 0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	33.4	24.7	32.0	24.1	30.5	23.2	26.1	31.3	25.2	30.1	24.2	28.9	5.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.2	29.4	23.6	19.1	28.4	22.7	18.0	27.4	82.3	31.4	78.1	30.1	74.4	28.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.9	(b) 10.7	(c) 9.3	(e) -24.4	(f) 35.8	(g) 3.3	(h) 2.2	(i) -26.7	(j) 37.4	(k) -28.7	(l) 38.7	(m) -30.5	(n) 40.0	(o) -32.9	(p) 41.6		
(4) 11.9	(5) 10.7	(6) 9.3	(7) -24.7	(8) 27.9	(9) 3.1	(10) 0.9	(11) -27.0	(12) 28.6	(13) -28.8	(14) 29.1	(15) -30.6	(16) 29.6	(17) -32.9	(18) 30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-4.9	-2.6	4.1	10.8	16.7	22.0	24.0	22.9	18.8	11.8	4.5	-2.0
	DBStd	11.41	6.97	7.00	6.52	5.43	4.64	3.43	3.23	3.03	4.51	5.37	5.84	6.44
	HDD10.0	1677	463	354	204	54	4	0	0	0	1	42	181	374
	HDD18.3	3386	721	585	441	232	88	9	1	3	49	212	414	630
	CDD10.0	1886	0	2	22	78	211	359	435	398	264	98	17	2
	CDD18.3	556	0	0	1	6	37	118	178	143	63	10	0	0
	CDH23.3	5007	0	1	7	73	316	1025	1731	1211	558	84	2	0
	CDH26.7	1741	0	0	1	14	76	344	689	419	182	17	0	0
Wind	WSAvg	4.7	5.3	5.3	5.5	5.6	4.9	4.2	3.6	3.4	3.9	4.6	5.1	5.0
Precipitation	PrecAvg	901	27	27	60	91	109	112	111	107	102	66	51	34
	PrecMax	1385	83	96	129	177	240	284	284	248	347	168	139	93
	PrecMin	571	3	4	11	27	18	20	20	10	12	1	2	2
	PrecStd	189	18	18	38	40	54	60	62	60	65	40	36	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.7	24.2	28.8	31.6	33.4	36.4	34.9	32.9	28.9	22.6	16.5
		MCWB	8.7	11.7	16.6	18.1	22.7	24.6	25.7	23.8	23.3	20.5	14.8	13.5
	2%	DB	10.0	13.8	20.8	25.5	28.7	32.0	33.9	32.4	30.9	25.8	18.9	12.6
		MCWB	6.2	9.1	14.4	16.7	20.6	23.8	25.2	24.4	22.6	18.1	13.2	9.4
	5%	DB	7.0	10.1	17.4	22.8	26.8	30.4	32.3	30.7	28.8	23.0	16.0	9.3
		MCWB	4.3	6.5	12.4	15.1	19.1	22.9	24.8	23.7	21.3	16.7	11.5	6.5
	10%	DB	3.9	6.8	14.1	20.2	24.9	28.7	30.5	29.0	26.7	20.7	13.4	6.5
		MCWB	1.9	4.2	9.5	13.6	18.2	22.2	23.7	22.8	20.0	15.9	9.6	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	13.2	17.7	19.9	23.6	26.4	27.5	26.9	24.9	22.1	16.5	14.4
		MCDB	12.4	16.5	22.7	26.2	29.3	31.4	33.0	31.4	30.8	26.6	20.8	15.9
	2%	WB	6.6	9.5	15.4	17.8	22.0	25.0	26.4	25.7	23.7	20.0	14.3	10.2
		MCDB	9.4	13.4	18.9	23.5	27.4	30.1	32.1	30.4	28.8	23.4	17.9	11.9
	5%	WB	4.2	6.8	12.9	16.2	20.7	24.0	25.5	24.7	22.6	18.1	12.2	6.6
		MCDB	7.0	10.1	17.0	21.5	25.1	28.8	30.7	29.2	27.0	21.8	15.5	9.0
	10%	WB	2.1	4.1	9.9	14.4	19.4	22.9	24.6	23.7	21.3	16.4	9.9	4.2
		MCDB	3.8	6.7	13.8	19.2	23.4	27.4	29.2	27.8	25.3	20.2	13.3	6.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.2	10.6	11.7	10.8	10.4	10.6	10.8	12.1	11.3	10.1	8.5
		MCDBR	12.6	13.5	14.7	15.3	12.6	11.8	11.8	12.4	13.3	13.6	13.6	11.9
		MCWBR	9.5	9.4	9.0	8.7	6.5	5.8	5.2	5.9	6.1	7.5	9.0	9.1
	5% WB	MCDBR	12.1	12.6	13.2	13.3	11.2	10.5	10.7	10.7	11.5	11.3	12.3	11.0
		MCWBR	9.4	9.2	8.6	8.5	6.5	6.1	5.5	6.0	5.9	7.5	9.0	9.1
Clear-Sky Solar Irradiance	taub	0.293	0.309	0.335	0.387	0.413	0.420	0.438	0.420	0.387	0.342	0.299	0.290	
	taud	2.417	2.371	2.390	2.261	2.255	2.281	2.219	2.287	2.331	2.458	2.507	2.446	
	Ebn at Noon	864	903	915	884	865	857	836	842	847	855	855	839	
	Edh at Noon	79	97	107	131	135	132	139	126	113	89	73	71	
All-Sky Solar Radiation	RadAvg	1.90	2.88	3.77	4.75	5.57	6.18	6.21	5.43	4.65	3.10	2.10	1.60	
	RadStd	0.19	0.22	0.32	0.35	0.40	0.62	0.45	0.32	0.34	0.35	0.18	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page